

PART 1: INSTALLATION

- 1. Install Maltego Community Edition.**
- 2. Create an account and log in.**
- 3. Explore the following sections:**
 - **Graph**
 - **Entity Palette**
 - **Transform Hub**
 - **Property View**

Take screenshots.

PART 2: DOMAIN INVESTIGATION

Choose any company or organization.

Examples:

- **Google**
- **Microsoft**
- **Amazon**
- **OpenAI**

Tasks:

- 1. Add a Domain Entity.**
- 2. Run available transforms.**
- 3. Collect information about:**
 - **Domain Names**
 - **Subdomains**
 - **IP Addresses**
 - **Email Addresses**
 - **Related Websites**

Take screenshots of the generated graph.

PART 3: EMAIL & COMPANY RESEARCH

Find and document:

- 1. Official Website**
- 2. Company Headquarters**
- 3. Social Media Accounts**
- 4. Public Contact Information**
- 5. Technologies used by the organization (if available)**

PART 4: VISUAL ANALYSIS

Create a relationship diagram showing:

Company



Website



Subdomains



IP Addresses



Public Information

Explain the relationships discovered.

ASSIGNMENT

Prepare a report containing:

- 1. Introduction to OSINT**
- 2. Introduction to Maltego**
- 3. Screenshots of Maltego Graphs**
- 4. Information Collected**
- 5. Relationship Diagram**
- 6. Security Risks of Public Information Exposure**
- 7. Recommendations for Organizations**

Research a company and answer:

- 1. How many domains are associated with the company?**
- 2. What public information is exposed?**
- 3. What sensitive information should organizations avoid exposing publicly?**

SUBMISSION

PDF or Word Document

1. Introduction to OSINT

Open Source Intelligence (OSINT) is the process of collecting and analyzing publicly available information from different online sources such as websites, search engines, social media, domain databases, and public records.

OSINT is widely used in cybersecurity for gathering information about organizations, identifying vulnerabilities, and understanding digital footprints. It helps security professionals discover what information is publicly exposed.

Uses of OSINT

- Threat intelligence
- Cybersecurity investigations
- Digital forensics
- Social engineering analysis
- Background checks
- Vulnerability assessment

2. Introduction to Maltego

Maltego is a powerful OSINT and graphical link analysis tool used to collect and visualize relationships between different entities.

It helps in:

- Domain investigation
- Subdomain enumeration
- IP tracing
- Email discovery
- WHOIS lookup
- Social media intelligence

Maltego displays data in graph form, making relationships easy to understand.

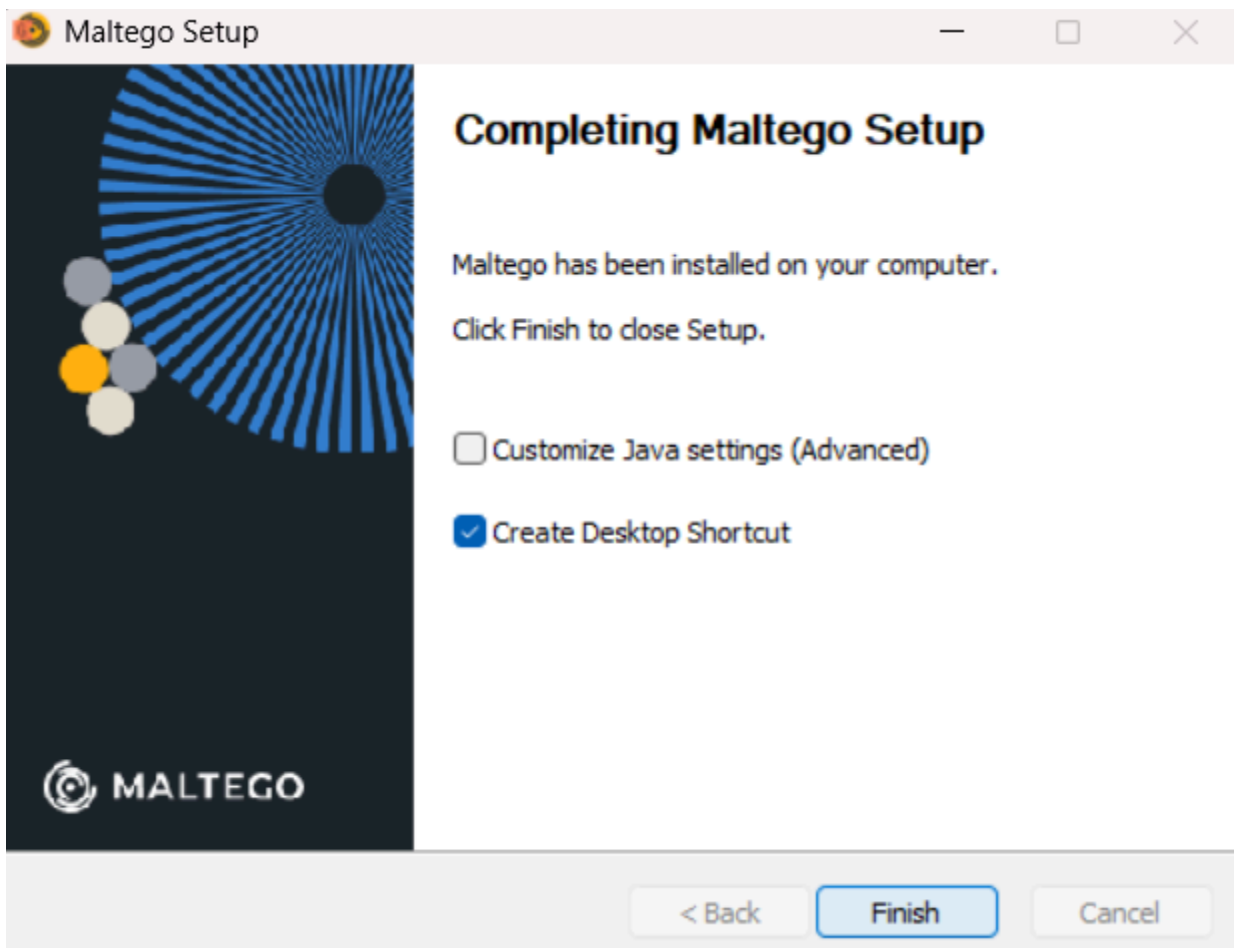
3. Installation of Maltego

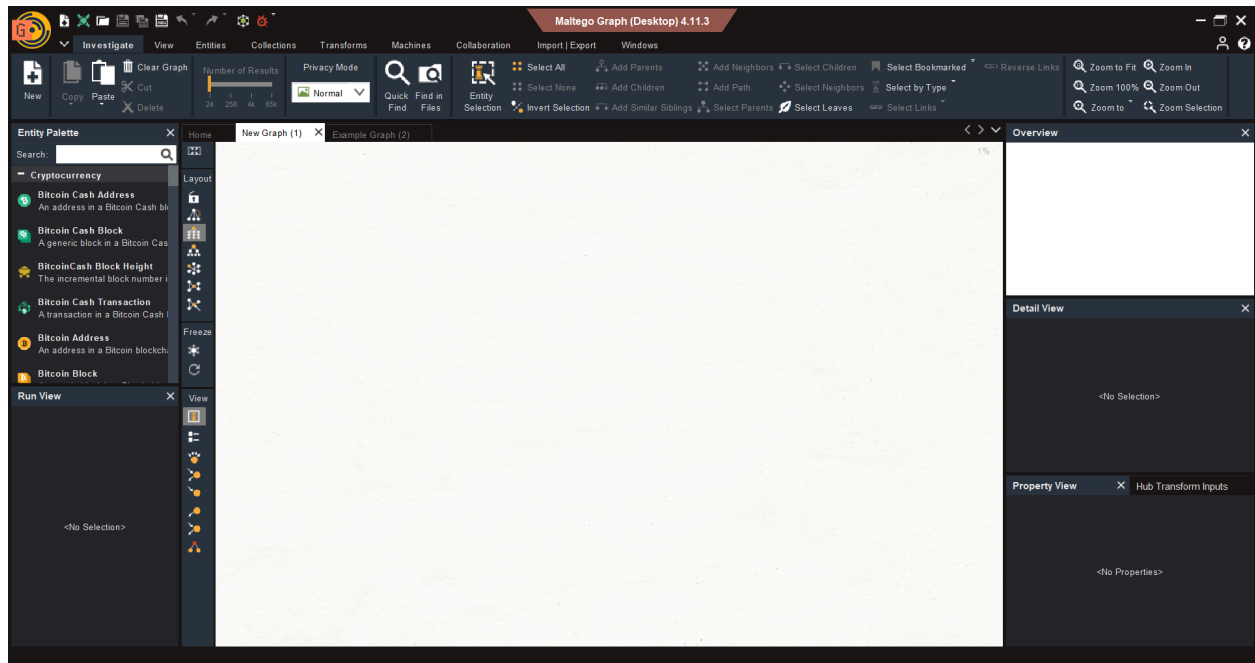
Steps Performed:

1. Downloaded and installed **Maltego**
2. Created Maltego account
3. Logged in successfully
4. Completed profile setup
5. Explored:
 - Graph
 - Entity Palette
 - Transform Hub
 - Property View

Screenshot:

Screenshot 1: Maltego Installation Completed





4. Exploring Maltego Interface

A. Graph

The graph is the main workspace where all entities and relationships are displayed.

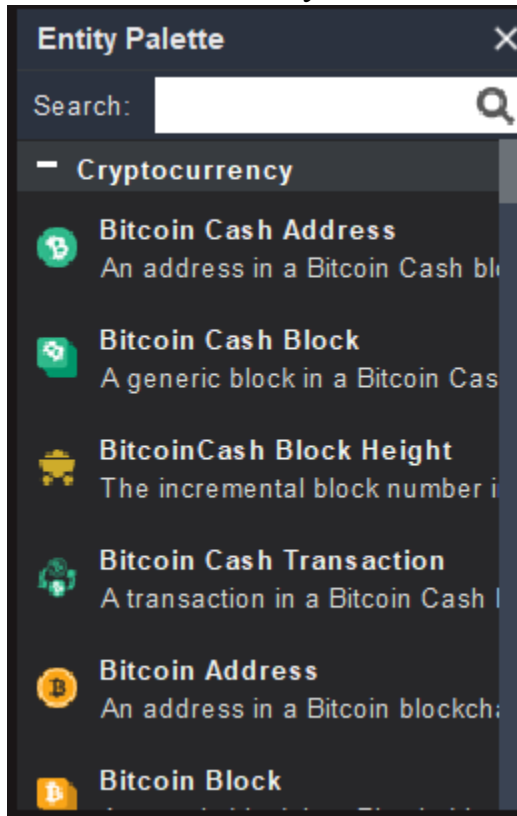
Screenshot 2: Blank Graph Workspace



B. Entity Palette

Entity Palette contains all available entities such as Domain, DNS Name, Website, Email, and Company.

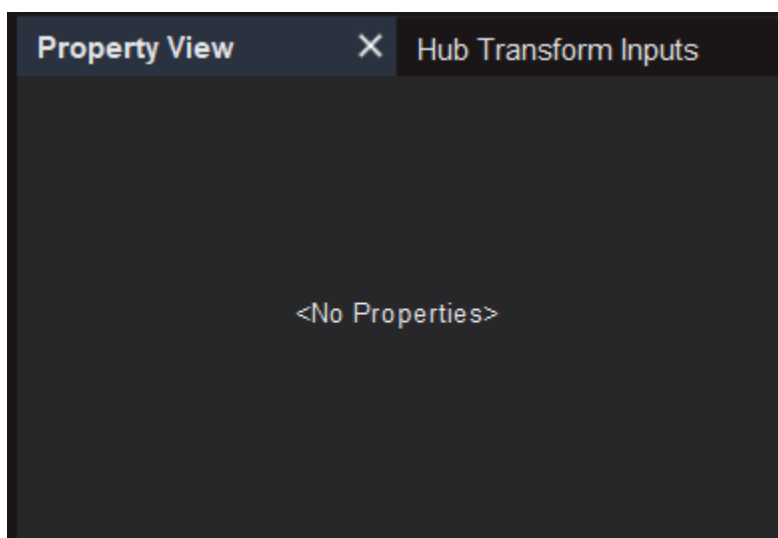
Screenshot 3: Entity Palette



C. Property View

Property View displays detailed information about selected entities.

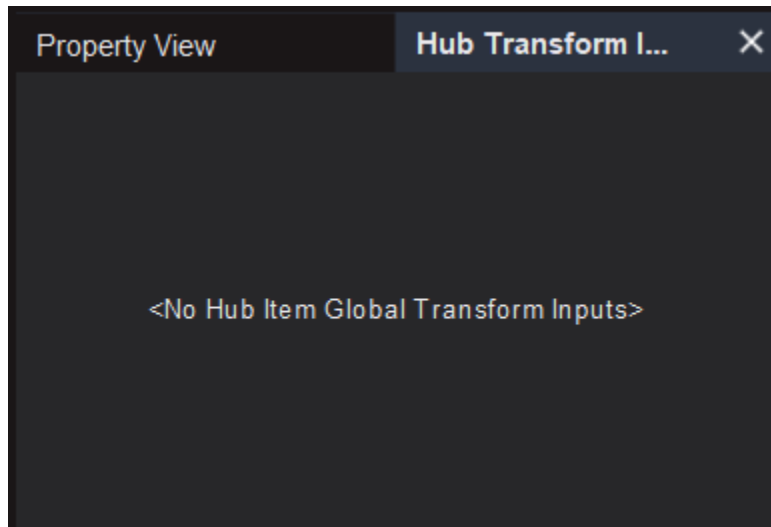
Screenshot 4: Property View



D. Transform Hub

Transform Hub allows us to run searches (transforms) to collect related information.

Screenshot 5: Hub Transform Inputs



5. Domain Investigation

Selected Company

OpenAI

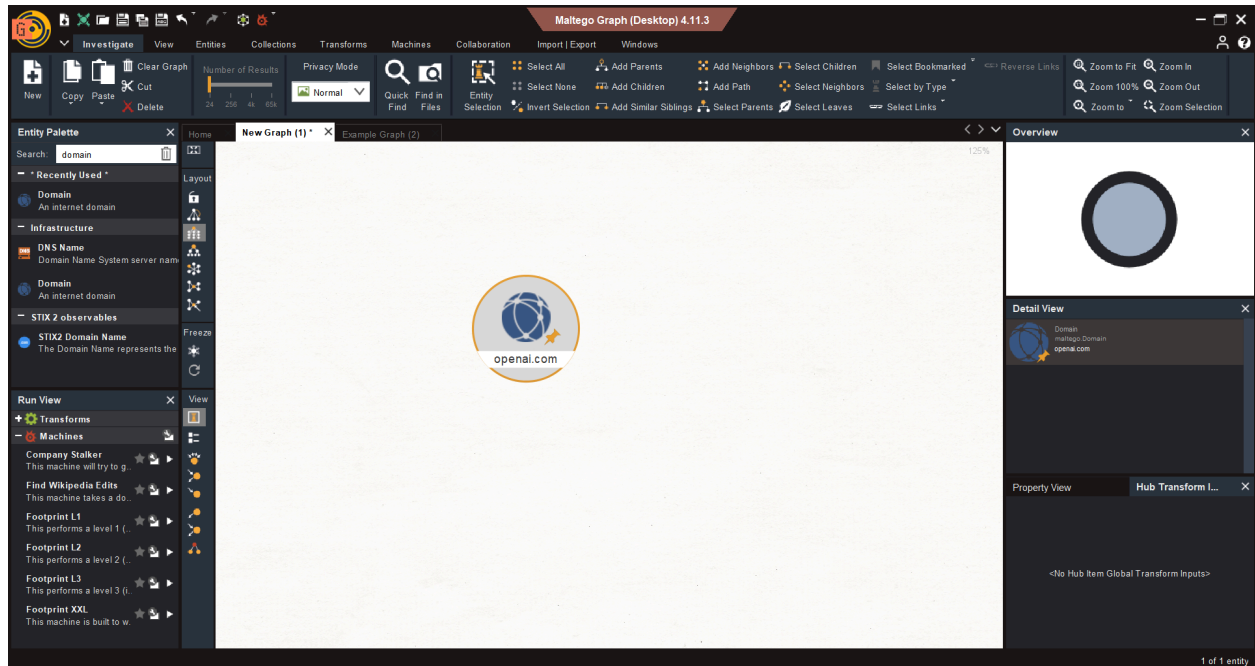
Domain Added

The domain selected for investigation was:

openai.com

A domain entity was added in Maltego graph.

Screenshot 6: Domain Added



6. Transforms Performed

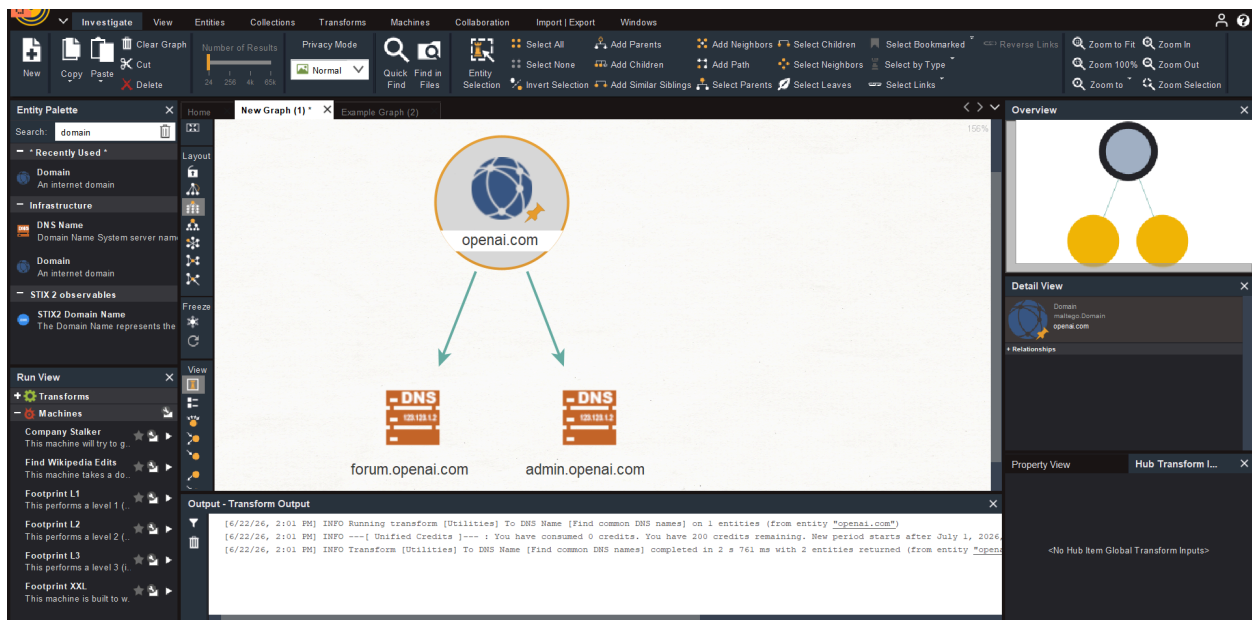
A. DNS Name Transform

The DNS transform was used to discover subdomains associated with the domain.

Subdomains found:

- admin.openai.com
- forum.openai.com

Screenshot 7: DNS Transform Results

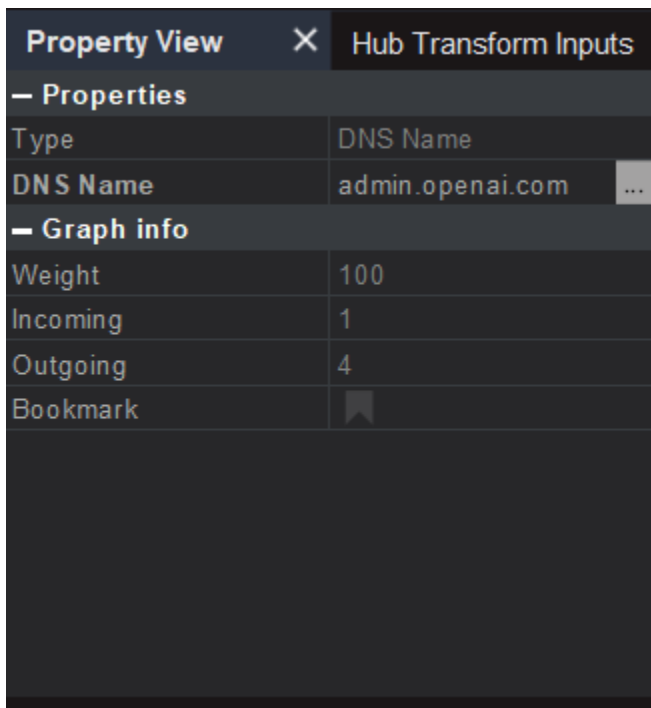


B. Entity Property Analysis

After selecting the discovered subdomain `admin.openai.com`, the Property View displayed the entity details such as type, DNS name, incoming and outgoing connections.

Observed details:

- Type: DNS Name
- DNS Name: `admin.openai.com`
- Weight: 100
- Incoming: 1
- Outgoing: 4



The screenshot shows a dark-themed interface with a 'Property View' tab and a 'Hub Transform Inputs' tab. The 'Properties' section lists 'Type' as 'DNS Name' and 'DNS Name' as 'admin.openai.com'. The 'Graph info' section lists 'Weight' as 100, 'Incoming' as 1, 'Outgoing' as 4, and a 'Bookmark' icon.

Property View		Hub Transform Inputs
— Properties		
Type	DNS Name	
DNS Name	admin.openai.com	...
— Graph info		
Weight	100	
Incoming	1	
Outgoing	4	
Bookmark		

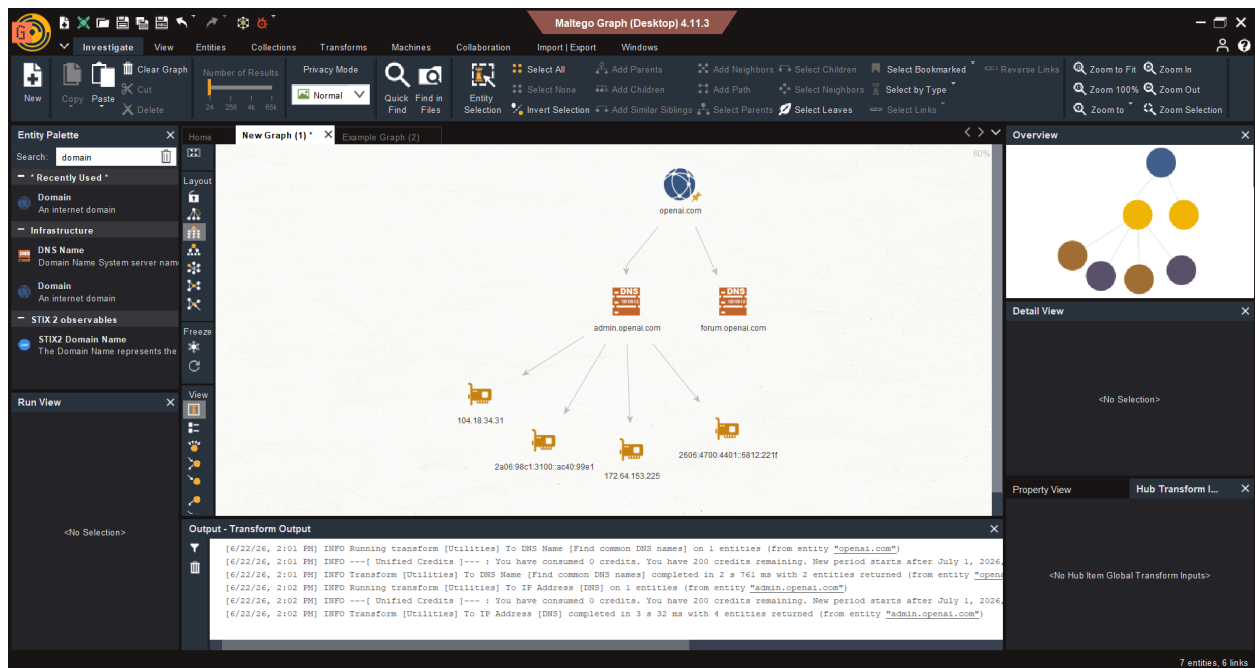
C. IP Address Transform

The IP transform was used on subdomains to identify associated server IPs.

IP Addresses found:

- 104.18.34.31
- 172.64.153.225
- 2a06:98c1:3100::ac40:99e1
- 2606:4700:4401::6812:221f

Screenshot 8: IP Transform Results



C. Website Transform

The website transform revealed:

- www.openai.com

Screenshot 9: Website Transform

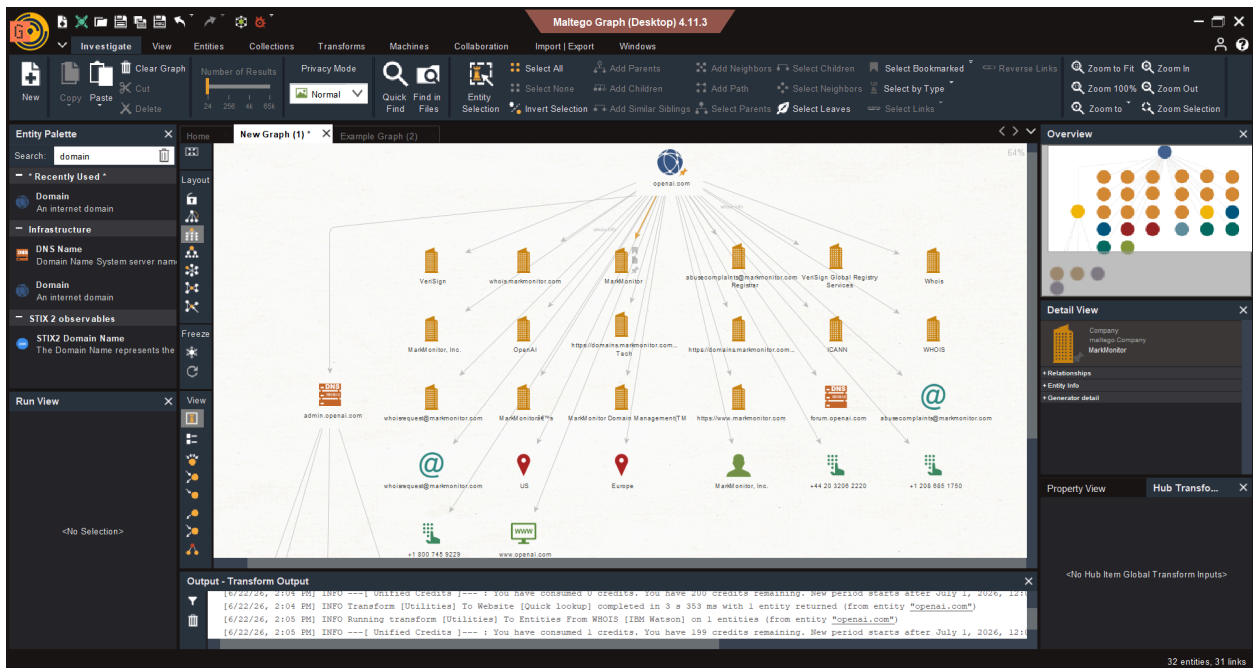


D. WHOIS Transform

WHOIS records were collected and revealed:

- Registrar details
- Contact emails
- Phone numbers
- Company details
- Regional information

Screenshot 10: WHOIS Graph



7. Information Collected

Category	Information Found
Main Domain	openai.com
Subdomains	admin.openai.com, forum.openai.com
IP Addresses	4
Website	www.openai.com
WHOIS Data	Available
Contact Emails	Found
Phone Numbers	Found

Email Addresses Found:

- whoisrequest@markmonitor.com
- abusecomplaints@markmonitor.com

8. Company Research

Official Website:

[OpenAI](#)

Headquarters

OpenAI Headquarters

Social Media Accounts

- [OpenAI X \(Twitter\)](#)
- [OpenAI LinkedIn](#)
- OpenAI YouTube

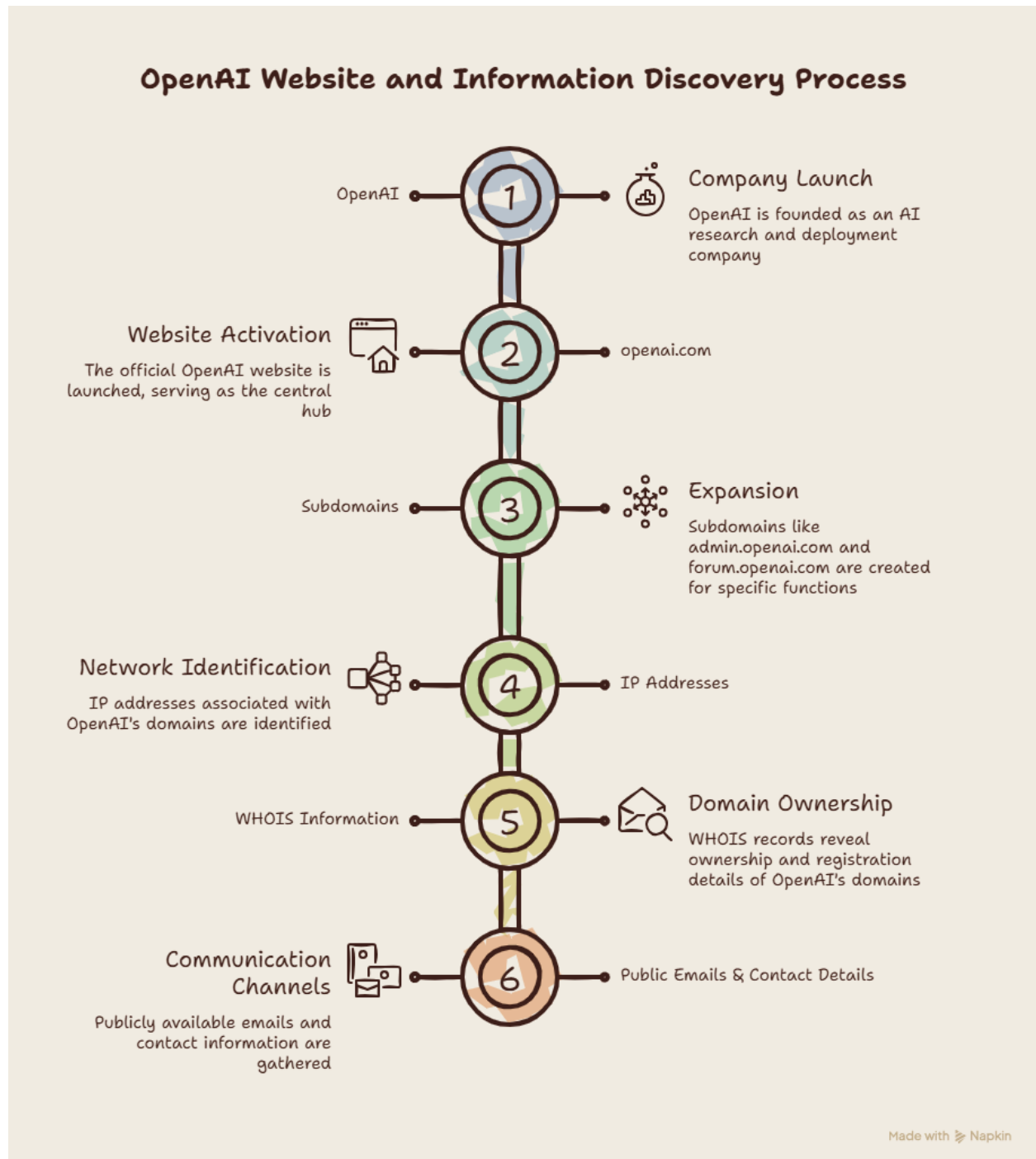
Public Contact Information

- WHOIS contact details
- Public support emails
- Phone numbers

Technologies Used

- Cloudflare
- Web Hosting Infrastructure
- DNS Services
- Security Certificates

9. Relationship Diagram



10. Relationship Analysis

The investigation started with the company **OpenAI** and its domain **openai.com**. The domain was linked to multiple subdomains which provide different services. These subdomains were connected to multiple IP addresses, showing the hosting infrastructure.

WHOIS information provided registrar details, contact information, and ownership records. This investigation shows how much information is publicly available about an organization.

11. Security Risks of Public Information Exposure

Public information exposure can lead to:

1. Phishing attacks using public emails
2. Infrastructure mapping by attackers
3. Subdomain enumeration
4. Social engineering attacks
5. Targeted cyberattacks
6. Information leakage
7. Attack surface expansion

12. Recommendations for Organizations

Organizations should:

1. Hide unnecessary subdomains
2. Limit public exposure of emails
3. Protect admin panels
4. Regularly audit OSINT exposure
5. Use firewalls and WAF
6. Monitor DNS records
7. Use email obfuscation
8. Restrict sensitive data exposure

13. Assignment Questions

Q1. How many domains are associated with the company?

Main domain found: **1**

Subdomains found: **2**

Total related domains discovered: **3**

Q2. What public information is exposed?

The following public information was exposed:

- Domain records
- DNS names
- IP addresses
- Registrar information
- Contact emails
- Phone numbers
- Website URLs
- Company data

Q3. What sensitive information should organizations avoid exposing publicly?

Organizations should avoid exposing:

- Internal admin portals
- Employee personal emails
- Internal IP addresses
- API endpoints
- Test environments
- Backup servers
- Internal documentation
- Security configurations

14. Conclusion

This experiment demonstrated how OSINT tools like **Maltego** can gather publicly available information about an organization. The investigation revealed multiple subdomains, IP addresses, and WHOIS information related to OpenAI.

This shows the importance of managing public exposure and reducing unnecessary information leakage to improve cybersecurity.